

Artemisia dracunculus L. (tarragon): a critical review of its traditional use, chemical composition, pharmacology, and safety.

Artemisia dracunculus L. (tarragon) has a long history of use as a spice and remedy. Two well-described "cultivars" (Russian and French) are used widely and differ in ploidy level, morphology, and chemistry. Key biologically active secondary metabolites are essential oils (0.15-3.1%), coumarins (>1%), flavonoids, and phenolcarboxylic acids. In vivo studies mainly in rodents, particularly from Russian sources, highlight potential anti-inflammatory, hepatoprotective, and antihyperglycemic effects. Despite concerns about the toxic effects of two of its main constituents, estragole (up to 82%) and methyleugenol (up to 39%), no acute toxicity or mutagenic activity has been reported at doses relevant for human consumption. Water extracts of *A. dracunculus* contain very low amounts of estragole and methyleugenol and, therefore, are considered to pose a very limited risk. Overall, a stronger focus on clinical studies and precise taxonomic and phytochemical definition of the source material will be essential for future research efforts.